MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Quiz 1 · Week 1: limits (Sections 1.5–1.7)** **10**

15 minutes. 10 points. No notes, no calculators, no phones. Write your answers in the spaces provided and show all your work. Write "DNE" if a limit does not exist.

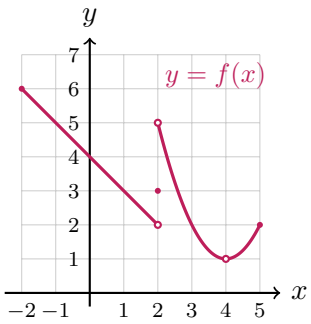
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1	2	3	Total

Problem 1 *Reading a graph.* **[4 points]**

The graph of a function f is shown. Find each of the following.



(a) $\lim_{x\to 2^-} f(x) =$ _____

(b) $\lim_{x\to 2^+} f(x) =$ _____

(c) $\lim_{x\to 2} f(x) =$ _____

(d) $\lim_{x\to 4} f(x) =$ _____

Problem 2 *From the lectures.* **[2 points]**

(Question 5 of Lecture 2, word for word.) Circle one letter, and show the calculation that gives it.

$$\lim_{x\to 0} x \sin\frac{1}{x} = ?$$

A. 0 **B.** 1

C. Does not exist — the function oscillates. **D.** ∞

Calculation:

Problem 3 *An ε–δ proof.* **[4 points]**

Prove from the definition that $\lim_{x\to 3}(5x - 4) = 11$. Scratch work on the left, the proof (written forward) on the right.

Scratch work — find δ

Proof